

Soft Landing and Inflation Scares

Heterogeneous-expectation New Keynesian (HENK) dynamics
Seminar notes on Bullard, Grimaud, Salle and Vermandel (2026)

Open-source notes on the Tinbergen DP / JME model
not the authors' replication package

`bullard-soft-landing/` in MacroModellerBot

September 2026

Source paper: Bullard, Grimaud, Salle and Vermandel (2026), Tinbergen Institute DP 2025-001
("Soft Landing and Inflation Scares"); JME 157 (2026), 103871.

Outline

- 1 Literature
- 2 The model
- 3 Contributions
- 4 FIRE vs. SL and the soft landing
- 5 What the repo prototype achieves
- 6 Simplifying assumptions
- 7 Extensions

Two tracks, in that order: the *paper* (literature, HENK, claims), then what this repo's Python/Dynare *prototype* actually does.

Why a soft landing?

Core question Bullard, Grimaud, Salle and Vermandel (2026)

Why did the post-pandemic U.S. inflation surge end in a *soft landing*—disinflation without a major recession—unlike Volcker-era disinflation?

- Stationary FIRE New Keynesian models mean-revert after a transitory cost-push; they do not generate a *distribution* of long-run inflation beliefs that can lose its anchor to the target.
- The paper asks whether the **timing** and **strength** of the Fed's reaction matter when those long-run beliefs are **heterogeneous**.
- That unanchoring is an **inflation scare** in the sense of Goodfriend (1993): a rise in perceived trend inflation that raises the real cost of subsequent disinflation.

Soft landing vs. inflation-scare literature

- **Goodfriend (1993).** Inflation scares are shifts in long-run inflation beliefs that force the central bank to choose between validating the scare and inflicting a recession. The 1980s disinflations are the classic case.
- **Post-COVID debate.** Was 2021–23 a transitory relative-price shock (FIRE-like), or a scare in which the distribution of trend-inflation beliefs moved? Bullard, Grimaud, Salle and Vermandel (2026) take the second view, disciplined by SPF forecast data as well as macros.
- **Policy timing.** The abstract's claim is that *when* the Fed tightened matters more than how large a later hike would have been for containing the scare (WP §4 counterfactuals).
- **This talk.** Paper first (HENK block, estimated claims), then an honest status report on the open-source prototype—not the authors' Table 3 paths.

FIRE vs. learning / heterogeneous expectations

- **FIRE NK.** A single rational forecast; belief wedge $\phi_t \equiv 0$. Cost-push inflation is temporary once the shock decays. Nested as the paper's homogeneous-belief special case.
- **Adaptive / statistical learning** (Evans and Honkapohja 2001; Hommes 2021 survey): agents estimate perceived laws of motion; mean beliefs can drift.
- **Social learning (SL)** (Grimaud, Salle and Vermandel, JEDC 2025, vol. 172): a finite population copies better-performing inflation intercepts (news $\rightarrow$ fitness $\rightarrow$ tournament). Dispersion is an *object*, not a residual. Authors solve HENK with this Dynare toolbox (Mendeley fzm3vkt66).
- **HENK restriction.** Only long-run *inflation* beliefs are non-FIRE; output-gap expectations in the IS curve remain model-consistent (internal rationality). Same SL family: Arifovic, Grimaud, Salle and Vermandel (*JMCB*) put CB announcements into the tournament as an anchoring device.

Not in the estimated HENK block—same post-COVID unanchoring queue:

- **Malmendier and Nagel** (NBER w35395). Stable long-run surveys can look “anchored” under experience learning alone, so they may understate de-anchoring after another high-persistence bout.
- **Gust, Herbst and López-Salido** (FEDS 2026-055). Firms endogenously lengthen planning horizons after large persistent shocks, raising inflation sensitivity and uncertainty.
- **Minton and Monnery** (FEDS 2026-053). Firms forecast own marginal costs, overreact to those and underreact to aggregates until costs move: the NKPC steepens and is less forward-looking; supply shocks inflate more; long-horizon forward guidance loses bite.

Contrast, not synthesis: Kaplan–Miyahara (NBER WP 35400) and Angeletos, Lian, Wolf and Zhang (NBER WP 35642) stress *balance-sheet* heterogeneity; HENK stresses *belief* heterogeneity in an otherwise RANK NK.

HENK at a glance

- Stylized NK economy: NKPC + Euler/IS + Taylor rule with smoothing.
- Population of J agents (J even), identical except for idiosyncratic long-run inflation intercepts $a_{j,t}$.
- Aggregate SL inflation forecast = cross-sectional average of individual forecasts.
- **Only departure from RE:** information friction on long-run inflation via social learning. Output-gap leads stay rational.
- Hats = deviations from steady state / inflation target. **Notation:** ϕ_t is the mean intercept (belief wedge); ϕ_π, ϕ_y in the Taylor rule are reaction coefficients.

Belief wedge (WP eq. 8)

$$E_t^{\text{SL}}(\hat{\pi}_{t+1}) = \phi_t + E_t(\hat{\pi}_{t+1}), \quad \phi_t \equiv \frac{1}{J} \sum_{j=1}^J a_{j,t}.$$

Nested FIRE: $\phi_t = 0$ for all $t \Rightarrow E_t^{\text{SL}} = E_t$.

Aggregate block (WP eqs. 1–3)

$$\hat{\pi}_t = \kappa \hat{y}_t + \beta E_t^{\text{SL}}(\hat{\pi}_{t+1}) + u_t \quad (\text{NKPC})$$

$$\hat{y}_t = E_t(\hat{y}_{t+1}) - \sigma^{-1}(\hat{i}_t - E_t^{\text{SL}}(\hat{\pi}_{t+1})) + g_t \quad (\text{IS})$$

$$\hat{i}_t = \rho_i \hat{i}_{t-1} + (1 - \rho_i)(\phi_\pi \hat{\pi}_t + \phi_y \hat{y}_t) + v_t \quad (\text{Taylor})$$

- Split leads: inflation is SL; the output-gap lead is RE. Cost-push u_t and demand g_t are ARMA(1,1); monetary v_t is white noise (WP §2.4).
- Seminar-level reading: a scare ($\phi_t \uparrow$) acts like a persistent inflation shock in the NKPC *and* eases the ex-ante real rate in the IS curve—costly to unwind later.

Assumption 2 (WP)

$$E_t(\phi_{t+1}) = \phi_t \quad (\text{random-walk / martingale mean belief}).$$

- Individual perceived law of motion (WP eq. 6):

$$\hat{\pi}_t^{(j)} = a_{j,t} + P_{1,\bullet} \hat{z}_{t-1} + \tilde{Q}_{1,\bullet} E_t.$$

- The paper keeps the *entire* cross-section $\{a_{j,t}\}$ in a closed form, then maps it into MSV rows $P, \tilde{Q}$ via the Dynare SL toolbox (SL_get_pq).
- Those rows are **not** recovered from the WP text alone; the Python prototype uses a documented stand-in (later slide).

Three-step recursive process each period, $j = 1, \dots, J$:

- 1 **News / mutation** (Calvo-like probability μ):

$$m_{j,t} = a_{j,t-1} + 1_{\{\varpi_{j,t} \leq \mu\}} (\iota_{j,t} + \lambda_t).$$

- 2 **Fitness**: discounted squared forecast errors on inflation history (decay ρ).
- 3 **Tournament**: agents paired; both copy the more accurate intercept.

Aggregate update $\phi_t = \phi_{t-1} + S(\cdot)$ with S nonlinear (no closed form for S). Authors solve the coupled system with the GSV Dynare SL toolbox. J is even in the theory; Table 1 does not estimate it.

Shocks and estimation (paper)

- Demand: $g_t = \rho_g g_{t-1} + \varepsilon_t^g - \mu_g \varepsilon_{t-1}^g$.
- Cost-push: $u_t = \rho_u u_{t-1} + \varepsilon_t^u - \mu_u \varepsilon_{t-1}^u$.
- Monetary: $v_t = \varepsilon_t^v$.
- SL news: aggregate λ_t , idiosyncratic $\iota_{j,t}$.
- Bayesian estimation with an inversion filter (Cuba-Borda, Guerrieri, Iacoviello and Zhong 2019) on **1985Q1–2023Q4** U.S. macros *and* SPF inflation forecasts (WP §3). $\beta = 0.99$ calibrated.

Param.	Table 1 mean	Param.	Table 1 mean
κ	0.2032	ρ_ι	0.8179
σ	1.6993	ρ_u, μ_u	0.6328, 0.4887
ϕ_π	1.7131	μ (news)	0.4357
ϕ_y	0.1564	ρ (fitness)	0.7745

Use Table 1 posterior means (not the WP narrative aside $\phi_\pi \approx 1.82$). Estimation itself is **not** in this repo.

What the Tinbergen / JME paper adds

Relative to FIRE NK and to the inflation-scare literature:

- A **micro-founded SL** layer that can *generate* inflation scares: heterogeneous long-run intercepts, time-varying dispersion, possible unanchoring from target.
- A **closed-form** representation that keeps the full cross-section of beliefs, then Bayesian estimation on macros+SPF.
- Quantitative claim: **timing** of tightening dominates delayed **strength** for containing the scare (conditional on policy not being too weak).
- Historical narrative (WP §4): the Fed was behind the curve in 2021; earlier tightening could have cut the inflation peak without a recession; further delay risks entrenchment and larger output losses.
- Soft landing as successful (if late) scare management under HENK—not as FIRE-only mean reversion.

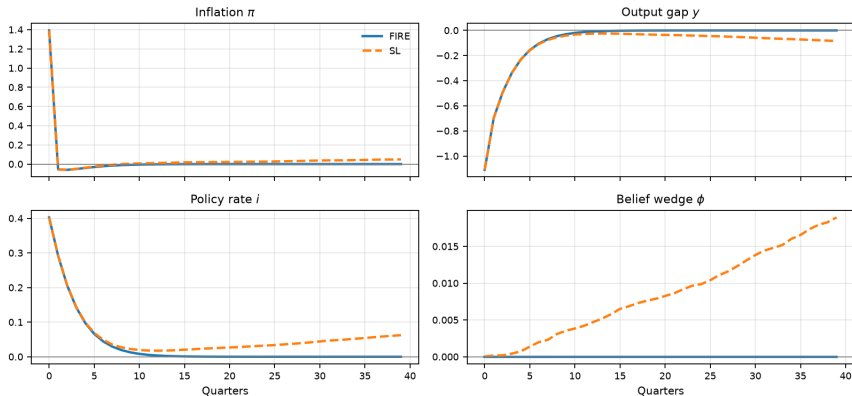
What this open-source prototype adds

Honest split vs. the published paper:

- **Runnable Python** FIRE nesting, finite- J SL micro block (WP eqs. 10–12), coupled HENK IRFs, and illustrative timing-vs-strength counterfactuals—no Matlab required.
- **Dynare scaffold** for the linear NK+ARMA block, aimed at the GSV SL toolbox once `TOOLBOX_PATH` is set.
- Fitness stand-in (provisional, not toolbox $P, \tilde{Q}$):
$$\mathbb{E}[\pi_s \mid m] = m + A_\pi i_{s-1} + D_\pi u_s + G_\pi g_s + B_\pi v_s.$$
- Extra solution approximation: each period SL uses *past* inflation, then a linear block takes ϕ_t as given (martingale $\mathbb{E}\phi' = \phi$). Not the toolbox's simultaneous SL+NK solution.
- **Does not** reproduce author Table 3 historical-shock paths, inversion-filter estimation, or SPF measurement equations. No official replication package was found.

Prototype IRF: cost-push, FIRE vs. SL

Cost-push IRF — FIRE vs social learning (prototype)



Python HENK prototype; unit cost-push impulse; Table 1 params; seed = 1, $J = 100$, ARMA(1,1) on; horizon 40.

[PROVISIONAL] not the authors' Dynare-toolbox IRFs.

Qualitative pattern in `output/fire_vs_sl_summary.txt` (and the figure):

- **FIRE** ($\phi \equiv 0$): inflation jumps on impact (unit shock: $\pi_0 \approx 1.39$), then undershoots and mean-reverts. Temporary cost-push, as in a standard 3-eq NK.
- **SL**: impact almost identical (beliefs have not had time to move). Thereafter ϕ_t drifts away from 0; inflation turns *positive again* around $t = 9$ while FIRE remains slightly below target.
- Mechanism: the wedge ϕ_t feeds the NKPC and the IS real rate; dispersed long-run beliefs amplify and persist the scare.
- Same structural parameters in both regimes—the extra persistence is *not* a more persistent u_t .

Timing vs. strength (prototype only)

Shared 12-quarter cost-push surge (0.01 innovations), identical SL seed, delay = 8 quarters (timing_counterfactual.py; NOTES.md STATUS):

Scenario	peak π	min y	max $ \phi $
Baseline (Table 1)	0.093	-0.169	0.037
Earlier ($\phi_\pi + 10\%$, $\rho_\iota \times 0.9$ from $t = 0$)	0.068	-0.158	0.034
Stronger delayed ($\phi_\pi + 10\%$ from $t = 8$)	0.075	-0.171	0.037

- Pattern is *consistent with* WP §4: earlier tightening beats delayed strength for peak inflation / scare containment.
- **Not** a claim of matching author Table 3 or estimated historical shocks. Peaks occur at the horizon end ($t = 79$); with a well-specified PLM, ϕ paths can be news-dominated.

Why this matters for 2021–22

- **Post-COVID surge.** A sequence of cost-push / relative-price shocks plus delayed liftoff is exactly the environment in which an inflation scare can develop if long-run beliefs are heterogeneous.
- **Anchoring is an object, and surveys can mislead.** Under FIRE, “transitory” is almost automatic. Under SL, anchoring is a *distribution* that policy timing can protect or lose. Malmendier–Nagel (w35395): stable long-run surveys can look anchored under experience learning alone.
- **Fed timing.** Paper: earlier 2021 tightening could have reduced the inflation peak without a recession; further delay would have unanchored expectations and raised output losses. Soft landing $\approx$ containing the scare late rather than never.
- **Complementary mechanisms** (not in HENK): endogenous horizons (Gust et al.) and own-cost NKPC (Minton–Monnery) also make large supply shocks more inflationary and weaken long-horizon forward guidance—another reason *current* timing beats distant promises.

Runs today (Python / NumPy; Table 1 params; unit impulses):

- Nested FIRE 3-eq monetary IRFs ($\phi_t = 0$).
- Finite- J SL micro: news, fitness, tournament (WP eqs. 10–12).
- Coupled HENK FIRE vs. SL IRFs with ARMA(1,1) u, g .
- Illustrative timing vs. strength under a shared surge.

Files: `re_nk_irf.py`, `social_learning.py`, `henk_sim.py` → `output/irf_*`, `timing_counterfactual.py` → `output/counterfactuals/`, `plot_costpush_irf.py`. FIRE paths nest `re_nk_irf` on the monetary A, B block.

Does not run: inversion-filter estimation, SPF observables, author Table 3, or toolbox $P, \tilde{Q}, R$ without Matlab + Mendeley zip.

- **Calibration:** Tinbergen Table 1 posterior means ($\kappa, \sigma, \phi_\pi, \phi_y, \rho_L$, ARMA, SL news/fitness) plus $\beta = 0.99$. Shock stds live in the Dynare .mod; Python IRFs default to *unit* impulses.
- **Dynare:** dynare/bullard_henk.mod is the FIRE/RE nesting of WP (1)–(3). Driver run_bullard_henk.m calls SL_get_pq then SL_chain *after* you set TOOLBOX_PATH to the unzipped Mendeley V3/Toolbox.
- **Not yet aligned:** Python C, D, G loadings vs. toolbox $P, \tilde{Q}, R$; inflation-only SL via $\text{mut_sd}_y = 0$ is provisional; no separate λ shock in toolbox options. Loadings D, G still use AR(1) expectations even when simulated paths are ARMA.
- **Blocked:** inversion-filter estimation, SPF observables, author Table 3, official replication files.

What is on, what is shut off

In the paper's baseline HENK:

- 3-equation NK block (no capital, no HANK, no open economy).
- Heterogeneity *only* in long-run inflation intercepts; output-gap expectations remain RE / model-consistent.
- Steady-state learning around a zero-deviation SS; hats vs. target.
- Assumption 2: martingale mean belief $\mathbb{E}\phi' = \phi$.

Extra prototype approximations (not author codes):

- Quasi-RE observer timing: SL sets ϕ_t from *past* inflation, then the linear block takes ϕ_t as given. Not a simultaneous nonlinear RE+SL fixed point.
- Delayed-strength loadings switch without agents anticipating the future rule. MA lag not in MSV loadings.

Shut off in both: own-cost-belief NKPC (Minton–Monnery); endogenous planning horizons (Gust et al.); Rotemberg microfoundations of κ (App. B).

Future developments that fit this codebase

- 1 **Dynare SL alignment.** Install GSV toolbox; set TOOLBOX_PATH; cross-check Python C , D , G against `mo.PP/QR/RR`; optional MA state in loadings.
- 2 **Phillips-block extensions.** Own-cost-belief NKPC (Minton–Monnery); App. B Rotemberg $\leftrightarrow \kappa$.
- 3 **Estimation stack** (blocked on data/codes). Inversion filter, SPF measurement (WP §3.1 / App. A), historical-shock Table 3.
- 4 **Companions in the literature queue.** Experience-based anchoring (Malmendier–Nagel w35395); endogenous finite horizons (Gust, Herbst and López-Salido); CB communication in the tournament (Arifovic et al.); UK/BoE timing analogue (NOTES.md).
- 5 **Later synthesis** (not this repo): HENK beliefs + HANK fiscal non-Ricardians for a fuller post-COVID accounting.

Takeaways

- Inflation scares are a *distributional* object: dispersed long-run inflation beliefs can persist after the cost-push dies.
- FIRE NK is nested ($\phi = 0$) and looks transitory; SL adds persistence through ϕ_t . Survey stability need not mean FIRE anchoring (Malmendier–Nagel).
- The paper's policy punchline is **timing over delayed strength**. The Python prototype is qualitatively consistent, not a numerical replica of Table 3.
- This repo is a transparent stepping stone: FIRE+SL IRFs, a Dynare scaffold, and documented gaps (toolbox, estimation, author codes).

Selected references. Bullard, Grimaud, Salle and Vermandel (2026), JME / Tinbergen DP 2025-001; Goodfriend (1993); Grimaud, Salle and Vermandel (JEDC 2025); Malmendier and Nagel (NBER w35395); Gust, Herbst and López-Salido (FEDS 2026-055); Minton and Monnery (FEDS 2026-053); Evans and Honkapohja (2001); Cuba-Borda et al. (2019); Arifovic, Grimaud, Salle and Vermandel (*JMCB*). Compile: `pdflatex henk_seminar.tex` from `bullard-soft-landing/beamer/`.